CROWD DYNAMICS IN THREE GOES · HANDBOOK

Go 1: Foundations

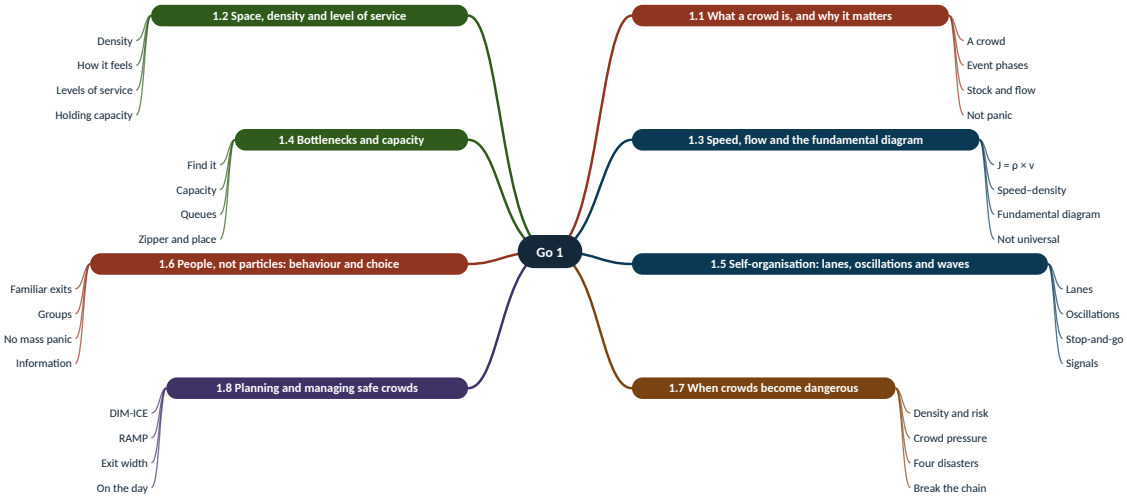
8 modules · about 4.8 hours of study

Companion to the website. Explorers, simulations, challenges and runnable code are online; this handbook keeps the text, worked numbers, mind maps and questions for reading offline.

Contents

1. **1.1 What a crowd is, and why it matters** · 30 min
   A crowd is many people sharing one space and often one purpose. Its behaviour comes from simple local rules, and its risks from too many people in too little space.
2. **1.2 Space, density and level of service** · 35 min
   Density is people per square metre; its inverse is the space each person has. Fruin's levels of service turn space per person into grades from A (free) to F (breakdown).
3. **1.3 Speed, flow and the fundamental diagram** · 40 min
   Flow = density × speed. As density rises, speed falls, so flow peaks (capacity) and then drops. This curve, the fundamental diagram, is the backbone of crowd science.
4. **1.4 Bottlenecks and capacity** · 35 min
   The narrowest point on a route sets the flow for the whole route. Capacity grows with width, about 1.3 to 1.9 people per metre per second in experiments, and queues grow when demand exceeds it.
5. **1.5 Self-organisation: lanes, oscillations and waves** · 30 min
   Without a leader, crowds form lanes in counterflow, take turns at narrow doors, and make stop-and-go waves at high density. Simple local rules create these patterns.
6. **1.6 People, not particles: behaviour and choice** · 35 min
   People choose routes, walk in groups, follow familiar paths and respond to information. Crowd behaviour is social, and 'panic' is rare. Good management works with that behaviour.
7. **1.7 When crowds become dangerous** · 40 min
   Risk rises steeply with density. Above about 5 to 6 people per m² people lose control of their movement; forces add up through bodies, and crowd turbulence and collapse can follow.
8. **1.8 Planning and managing safe crowds** · 40 min
   Safe events are designed, informed and managed in every phase. DIM-ICE and RAMP structure the risk analysis; flow rates and egress times size the exits.

Mind map of the go



1.1 What a crowd is, and why it matters

30 minutes · Big picture

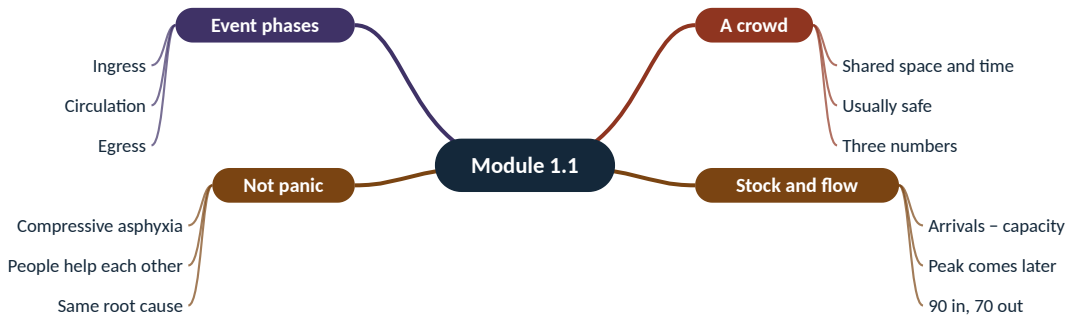
A crowd is many people sharing one space and often one purpose. Its behaviour comes from simple local rules, and its risks from too many people in too little space.

By the end you can

- say what makes a group of people a crowd, and name the three phases of an event
- explain why crowd behaviour is about space and flow, not panic
- read a queue as a stock that fills and empties
- name three famous crowd disasters and what they had in common

Overview

A crowd is a large number of people in one place at the same time, often sharing a purpose: watching a match, catching a train, or performing a pilgrimage. Crowds are usually safe, friendly and even pleasant. But they follow a simple physics. Every person needs space, and every route can carry only so many people each minute. When more people arrive than a space or a gate can pass, they wait, and the waiting crowd grows like water filling a bath. Most deadly crowd incidents were not caused by people panicking. They happened because too many people were packed into too little space, so that pressure built up and people could no longer move or breathe. In this course you will learn to see a crowd three ways: as people with goals, as a flow through space, and as a system that designers and managers can keep safe. We start with the big picture: what a crowd is, the three phases of an event, and why the numbers matter.



A crowd, in plain words

A crowd is a large number of people in one place at the same time. Some crowds share a purpose (a concert audience, pilgrims, commuters); others are just people passing (a busy street). Crowd scientists care about crowds because **most of the time they are safe and enjoyable, and occasionally they are deadly**, and the difference is usually predictable from numbers.

We describe a crowd with three linked quantities: **density** (people per square metre), **speed** (metres per second) and **flow** (people passing a line per second).

Three phases of every event

Every event has three phases: **ingress** (arriving), **circulation** (moving around inside) and **egress** (leaving). Each has its own peak. Egress is often the most intense, because everyone leaves at once at the end.

Crowd safety expert Keith Still built these phases into his DIM-ICE model, which you will meet in module 1.8.

A queue is a stock

Picture a bath. The tap is the arrival rate, the plug hole is the gate's capacity, and the water level is the number of people waiting. If the tap runs faster than the plug drains, the level rises. It keeps rising until arrivals fall below capacity, so **the biggest queue comes after the busiest arrival time**.

$$\text{change in people waiting} = \text{arrivals} - \text{departures}$$

A worked example: 90 people a minute arrive at a gate that passes 70. The queue grows by 20 people every minute.

TRY Try it in your head

A stadium gate passes 60 people a minute. For 30 minutes, 100 people a minute arrive. How many are waiting at the end? *(100 − 60) × 30 = 1,200.*

Crowd crushes are not panic

News reports often say a crowd 'panicked' or 'stampeded'. Research tells a different story. In most deadly incidents, people were packed so tightly that they could not move, breathe or escape: **compressive asphyxia**. People in danger mostly help each other (Drury, Cocking and Reicher 2009).

Examples you will meet again: the Hillsborough stadium disaster (1989, 97 deaths), the Love Parade in Duisburg (2010, 21 deaths), the Mina crush during the Hajj (2015, 769 deaths in the official Saudi count; independent counts exceed 2,000) and the Itaewon crowd crush in Seoul (2022, 159 deaths). In each, too many people reached too little space.

TRAP The word 'stampede'

'Stampede' suggests animals running wildly and blames the victims. Most crushes happen where people are barely moving. Prefer 'crowd crush' or 'crowd collapse'.

LAB A queue as a stock

Each step adds (arrivals − capacity) × 5 minutes to the queue, never below zero. Change capacity to 100 and run again.

```
minute   5: arrivals  20/min, waiting     0
...
minute  25: arrivals  90/min, waiting   400
minute  35: arrivals  20/min, waiting     0
```

Takeaways

- A crowd is described by density, speed and flow.
- Every event has ingress, circulation and egress phases, each with its own peak.
- A waiting crowd is a stock: it grows while arrivals exceed capacity, and peaks after the arrival peak.
- Deadly crowd incidents are about density and pressure, not panic.

Check yourself

1. A gate passes 50 people per minute and 80 arrive per minute for 10 minutes. How many are waiting at the end?
   (a) 30 · (b) 300 · (c) 800 · (d) 130
2. Which event phase is often the most intense?
   (a) Ingress · (b) Circulation · (c) Egress · (d) Set-up
3. What killed most victims of major crowd crushes?
   (a) Trampling during a panic · (b) Compressive asphyxia from very high density · (c) Heat · (d) Fire
4. The biggest queue usually forms…
   (a) Before arrivals peak · (b) Exactly when arrivals peak · (c) After arrivals peak · (d) At random

Answers: 1 (b) (80 − 50) × 10 = 300 people.; 2 (c) At the end everyone leaves at once, so egress flows are often the largest.; 3 (b) People were squeezed so tightly they could not breathe.; 4 (c) The queue keeps growing until arrivals fall below capacity, which is after the peak.

Words

- **Crowd**: A large number of people gathered in one place at one time, often sharing a purpose.
- **Ingress**: The arrival phase of an event.
- **Circulation**: Movement within a venue during an event.
- **Egress**: The departure phase of an event, normal or emergency.
- **Compressive asphyxia**: Inability to breathe because the chest is squeezed, the main cause of death in crowd crushes.
- **Stock**: A quantity that accumulates, such as the number of people waiting.

Reading

- Drury, Cocking and Reicher (2009): Everyone for themselves?: Evidence that people in emergencies cooperate rather than panic. https://doi.org/10.1348/014466608X357893
- Still: Introduction to Crowd Science (book site): The crowd science textbook behind DIM-ICE and RAMP analysis. https://www.gkstill.com/Support/Book.html
- UK HSE: Managing crowds safely (HSG154): A short official guide to planning for crowds at venues and events. https://www.hse.gov.uk/pubns/books/hsg154.htm

1.2 Space, density and level of service

35 minutes · Measure

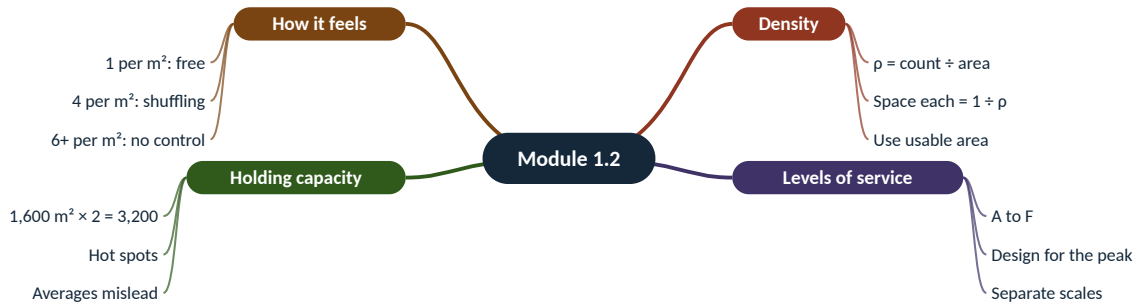
Density is people per square metre; its inverse is the space each person has. Fruin's levels of service turn space per person into grades from A (free) to F (breakdown).

By the end you can

- calculate density and space per person from a count and an area
- read a level of service from A to F
- explain why the same number of people can be fine in one place and dangerous in another
- estimate how many people fit in an area at a chosen comfort level

Overview

How full is a space? Crowd scientists answer with one number: density, the number of people per square metre. Count the people in an area and divide by the area. Turn it upside down and you get the space each person has. At one person per square metre you can walk freely. At two, you have to adjust your speed and path. At four, you are shuffling and brushing against others. At six or more, you are pressed by bodies on all sides and cannot move on your own. In the nineteen seventies the engineer John Fruin turned these experiences into levels of service, graded from A, free movement, to F, where flow breaks down. Planners still use them to design walkways, platforms and queues. In this module you will measure density, read levels of service, and estimate how many people a space can hold comfortably. Remember: the same thousand people are comfortable in a large square and dangerous in a narrow street.



Density, the key number

Density is the number of people per square metre, written ρ (rho). Count the people in a known area and divide:

$$\rho = \text{number of people} \div \text{area} \ (\text{people per m}^2)$$

Example: 120 people in a 60 m² lobby give ρ = 2 people per m². The **space per person** is the inverse: 60 ÷ 120 = 0.5 m² each.

TRY Quick check

A platform is 100 m long and 4 m wide, and 600 people wait on it. Density? *600 ÷ 400 = 1.5 per m², about 0.67 m² each: level E for a walkway, tolerable for waiting.*

What density feels like

Rough guide, for adults moving in one direction: **below 1 per m²** you walk freely; **about 2** you adjust speed and path; **about 4** you shuffle with frequent contact; **above about 5 to 6** you cannot move independently and pressure can build (Fruin 1993; Still 2014). Standing still, people tolerate higher densities than walking, which is why queues and viewing areas use different thresholds.

Levels of service

John Fruin (1971) graded walkways A to F by space per person, like grades on a report card. Designers choose a target grade for the busiest 15 minutes, often C or D for stations and B or C for comfortable public spaces. Separate scales exist for **queuing areas** and **stairs**, because standing and climbing need different space.

How many people fit?

Run the calculation backwards. A 2,000 m² square, of which 1,600 m² is usable, at a comfortable 2 people per m² holds 3,200 people. At 4 per m² it holds 6,400, but people can no longer move freely. Remember that density is never even: people gather near stages, exits and screens, so **average density hides dangerous hot spots**.

TRAP Average density

'The square was only half full' can still hide 6 per m² at the front. Always ask where the people were, not just how many.

LAB Density and level of service

Fruin's walkway thresholds in m² per person. Add your own lobby or platform to the list.

```
  30 people in  120 m2: 0.25 per m2, 4.00 m2 each, LOS A
 120 people in   60 m2: 2.00 per m2, 0.50 m2 each, LOS E
```

Takeaways

- Density ρ = people ÷ area; space per person = 1 ÷ ρ.
- Below about 1 per m² people walk freely; above about 5–6 per m² they cannot move independently.
- Fruin's levels of service grade space per person from A to F.
- Holding capacity = usable area × accepted density, but averages hide hot spots.

Check yourself

1. 200 people stand in a 50 m² area. What is the density?
   (a) 0.25 per m² · (b) 4 per m² · (c) 250 per m² · (d) 40 per m²
2. At 2 people per m², how much space does each person have?
   (a) 2 m² · (b) 0.5 m² · (c) 4 m² · (d) 0.2 m²

3. Fruin's level of service F means…
   (a) Free flow · (b) Flow breaks down, frequent contact · (c) Fast walking · (d) A fire exit
4. Why can an average density be misleading?
   (a) Densities are always even · (b) People cluster at hot spots like stage fronts and exits · (c) Cameras cannot count · (d) Density changes with weather

Answers: 1 (b) 200 ÷ 50 = 4 people per m².; 2 (b) Space per person = 1 ÷ 2 = 0.5 m².; 3 (b) F is under about 0.5 m² per person, where movement breaks down.; 4 (b) Crowds are uneven; danger is local.

Words

- **Density**: Number of people per square metre (ρ).
- **Space per person**: Area available to each person, the inverse of density, in m².
- **Level of service**: A grade (A to F) of crowd comfort based on space per person or flow (Fruin 1971).
- **Holding capacity**: The number of people an area can hold at an accepted density.
- **Usable area**: The area people can actually occupy, excluding obstacles.

Reading

- Fruin (1971): Designing for pedestrians, a level-of-service concept: The original level-of-service paper (library record). https://swov.nl/en/publicatie/designing-pedestrians-level-service-concept
- HSE: Managing crowds safely: Practical guidance on capacity and density at events. https://www.hse.gov.uk/pubns/books/hsg154.htm
- Johansson et al. (2008): From crowd dynamics to crowd safety: Measured densities above 6 per m² at the Jamarat Bridge. https://arxiv.org/abs/0810.4590

1.3 Speed, flow and the fundamental diagram

40 minutes · Core idea

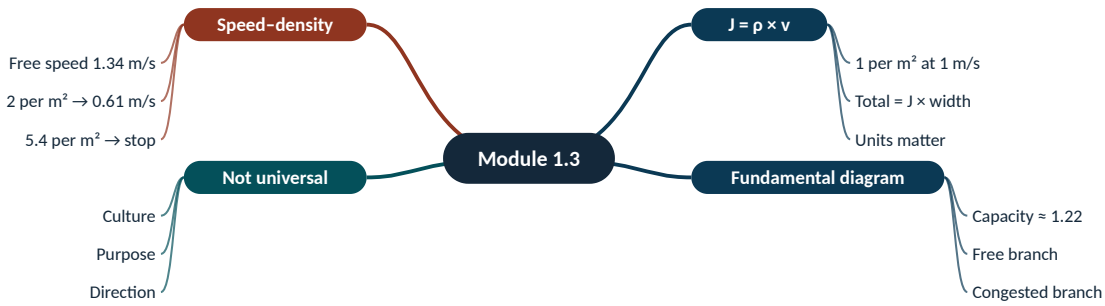
Flow = density × speed. As density rises, speed falls, so flow peaks (capacity) and then drops. This curve, the fundamental diagram, is the backbone of crowd science.

By the end you can

- use flow = density × speed
- sketch the fundamental diagram and mark capacity
- explain why adding people to a busy corridor can lower its flow
- use Weidmann's speed–density relation for a quick estimate

Overview

Imagine a corridor. With a few people, everyone walks at their own comfortable speed, about one point three metres per second. Add more people and two things happen. More people pass each second, because there are more of them, but each person slows down, because others are in the way. Flow, the number of people crossing a line each second per metre of width, is density times speed. At first density wins and flow rises. Then speed falls so fast that flow stops rising, reaches a maximum, and falls again. That maximum is the capacity of the corridor, about one point two people per metre per second in the classic curve by Ulrich Weidmann, reached at a density of nearly two people per square metre. The curve of flow against density is called the fundamental diagram. It is the single most important picture in crowd dynamics, and every model you meet later must reproduce it.



Flow = density × speed

Think of a corridor 1 m wide. If density is 1 person per m² and everyone walks at 1 m/s, one person crosses a line each second. The rule is:

$$J = \rho \times v \quad \text{(people per metre of width per second)}$$

For a corridor of width b, the total flow is **J × b**. Example: ρ = 2 per m², v = 0.6 m/s, b = 3 m gives J = 1.2 per m per s and a total of 3.6 people per second, about 216 per minute.

Speed falls as density rises

Alone, adults walk at about **1.34 m/s** on average (Weidmann 1993). With others around, they slow down. Weidmann summarised 25 studies with a formula:

$$v(\rho) = 1.34 \times [1 - \exp(-1.913 \times (1/\rho - 1/5.4))] \text{ m/s}$$

At 1 per m² it gives about 1.06 m/s, at 2 per m² about 0.61 m/s, and at 5.4 per m² zero: everyone stands still.

The fundamental diagram

Multiply speed by density and plot flow against density: the curve rises, peaks and falls. The peak is **capacity**: about **1.22 people per metre per second** at about **1.75 per m²** in Weidmann's curve. Left of the peak the flow is **free**; right of the peak it is **congested**, and adding people reduces flow.

This is why a corridor operated near capacity is fragile: a small disturbance can push it onto the congested branch, where it moves fewer people and density keeps rising.

MODEL Capacity in practice

A 4 m wide passage at 1.2 per m per s carries 4.8 people per second, about 290 per minute or 17,000 per hour. Guidance documents use similar or lower planning values for safety.

One curve does not fit all crowds

Measured diagrams differ. Single-file experiments in Jülich (Seyfried et al. 2005), crowds in India and Germany (Chattaraj, Seyfried and Chakroborty 2009) and pilgrims at the Jamarat Bridge (Helbing et al. 2007) all give different curves. Purpose matters too: commuters walk faster than shoppers, and two-way flows behave differently from one-way flows.

TRAP Borrowing a curve

Using one fundamental diagram everywhere can overestimate capacity for older or slower crowds, such as many pilgrims.

LAB Weidmann's fundamental diagram

Computes Weidmann's curve and finds its peak. Change v0 to 1.1 m/s for a slower crowd and see how capacity drops.

```
capacity 1.22 per m per s at 1.75 per m2, speed 0.70 m/s
```

Takeaways

- Specific flow J = density × speed; total flow = J × width.
- Speed falls with density, from about 1.34 m/s to zero near 5.4 per m² (Weidmann).
- Flow peaks at capacity (about 1.2 per m per s) then falls: the fundamental diagram.

- Fundamental diagrams vary with culture, purpose and direction of flow.

Check yourself

1. Density is 2 per m² and speed 0.6 m/s. What is the specific flow?
   (a) 0.3 per m per s · (b) 1.2 per m per s · (c) 2.6 per m per s · (d) 3.3 per m per s
2. On the congested branch of the fundamental diagram, adding people…
   (a) Raises flow · (b) Lowers flow · (c) Does nothing · (d) Doubles speed
3. Weidmann's capacity is about…
   (a) 0.5 per m per s · (b) 1.2 per m per s · (c) 5 per m per s · (d) 10 per m per s
4. Why measure the fundamental diagram for your own crowd?
   (a) It is identical everywhere · (b) It varies with culture, purpose, age and flow direction · (c) It is required by law everywhere · (d) To check the cameras

Answers: 1 (b) J = 2 × 0.6 = 1.2.; 2 (b) Speed drops faster than density rises.; 3 (b) About 1.22 at 1.75 per m².; 4 (b) Imported curves can mislead.

Words

- **Speed**: How fast people walk, in metres per second.
- **Specific flow**: People passing per metre of width per second (J = ρ × v).
- **Fundamental diagram**: The relation between density and flow (or speed) of a crowd.
- **Capacity**: The maximum flow a facility can carry.
- **Free speed**: The speed people walk when unhindered, about 1.34 m/s on average (Weidmann).
- **Jam density**: The density at which movement stops; 5.4 per m² in Weidmann's curve.

Reading

- Seyfried et al. (2005): The fundamental diagram of pedestrian movement revisited: Classic single-file experiment and a clear introduction to the fundamental diagram. https://arxiv.org/abs/physics/0506170
- Chattaraj, Seyfried and Chakroborty (2009): FD across cultures: Indian and German experiments give different fundamental diagrams. https://arxiv.org/abs/0903.0149
- Helbing, Johansson and Al-Abideen (2007): Dynamics of crowd disasters: The fundamental diagram measured at the Jamarat Bridge, at extreme densities. https://arxiv.org/abs/physics/0701203

1.4 Bottlenecks and capacity

35 minutes · Core idea

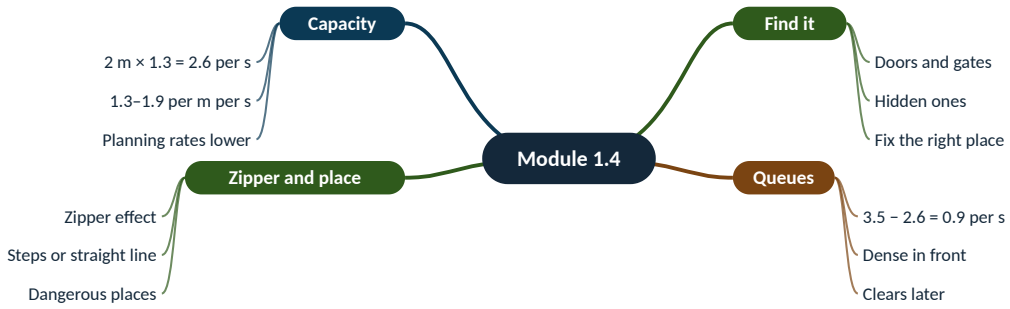
The narrowest point on a route sets the flow for the whole route. Capacity grows with width, about 1.3 to 1.9 people per metre per second in experiments, and queues grow when demand exceeds it.

By the end you can

- find the bottleneck on a route
- calculate a door's capacity from its width
- estimate how fast a queue grows and how long it takes to clear
- describe the zipper effect and why wider is not always proportionally better

Overview

Think of a wide corridor ending in a single door. However wide the corridor, everyone must squeeze through that door. The narrowest point on a route is the bottleneck, and its capacity sets the flow for the whole route. A good first estimate is simple: capacity equals the width of the bottleneck times a specific flow, typically between about one point three and one point nine people per metre per second in laboratory experiments. If people arrive faster than that, a queue forms in front of the bottleneck and grows by the difference every second. Inside a bottleneck, people often interleave like the teeth of a zipper, using the space efficiently. In this module you will find bottlenecks, calculate their capacity, and estimate how long a crowd takes to pass through. You will also see why a bottleneck in the wrong place, such as a narrowing at the bottom of a slope, can be dangerous.



Find the bottleneck

Walk the route from start to finish and look for the narrowest effective width: a door, a turnstile line, a stair, a gate in a fence, a street narrowed by stalls. The route can carry no more than that point allows. **Upgrading anything else does not help.**

Capacity from width

$$\text{capacity} = \text{width} \times \text{specific flow} \quad (\text{people per second})$$

Laboratory experiments found specific flows of about **1.3 to 1.9 people per metre per second** through bottlenecks (Seyfried et al. 2009). A 2 m door at 1.3 per m per s passes 2.6 people per second, or 156 per minute: 2,000 people take about 13 minutes.

Guidance for stadiums uses planning rates per minute instead, such as 82 people per metre per minute on level routes in the Green Guide (module 1.8).

TRY Your turn

A 1.2 m exit at 1.3 per m per s. How long for 500 people? *Capacity 1.56 per s, so 500 ÷ 1.56 ≈ 320 s, about 5.3 minutes.*

How queues grow

If 3.5 people arrive per second at that 2 m door, the queue grows by 3.5 − 2.6 = 0.9 people per second: 540 people after 10 minutes. The queue packs at high density right in front of the door, often 3 to 5 per m² or more.

Zippers, steps and bad places

Inside bottlenecks people form **overlapping lanes**, the zipper effect. Some experiments suggested capacity rises in steps as each new lane fits (Hoogendoorn and Daamen 2005); others found nearly linear growth (Seyfried et al. 2009). For design, linear growth is a reasonable first estimate.

Where a bottleneck sits matters: a narrowing at the end of a downhill slope, at a T-junction, or where two flows meet concentrates pressure. The Itaewon alley in 2022 was narrow, sloping and fed from both ends.

TRAP Doubling the width

Doubling width roughly doubles capacity only if the space before and after the bottleneck can supply and absorb the extra flow.

LAB Time to pass a bottleneck

Compares the time for 2,000 people at low and high specific flows.

```
width 1.0 m, Js 1.3: capacity 1.3/s, 25.6 min for 2000
```

Takeaways

- The bottleneck, the narrowest point, sets a route's capacity.
- Capacity ≈ width × specific flow (about 1.3 to 1.9 per m per s in experiments).
- Queues grow at demand minus capacity, and pack densely in front of the bottleneck.
- Where a bottleneck sits (slopes, junctions, meeting flows) can matter as much as its width.

Check yourself

1. A 3 m gate with specific flow 1.3 per m per s passes how many people per minute?
   (a) 39 · (b) 234 · (c) 390 · (d) 78

2. Widening a corridor upstream of a narrow door will…
   (a) Double the flow · (b) Not raise the route's capacity · (c) Halve the queue · (d) Remove the zipper effect
3. Demand is 5 per s and capacity 4 per s. After 5 minutes the queue is…
   (a) 60 · (b) 300 · (c) 1,200 · (d) 1,500
4. The 'zipper effect' describes…
   (a) People forming overlapping lanes in a bottleneck · (b) A broken gate · (c) Ticket checks · (d) Fast walkers overtaking

Answers: 1 (b) 3 × 1.3 = 3.9 per s × 60 = 234 per minute.; 2 (b) The door still limits the flow.; 3 (b) (5 − 4) × 300 s = 300 people.; 4 (a) Lanes interleave to use the width efficiently.

Words

- **Bottleneck**: The narrowest point on a route; it limits the route's flow.
- **Effective width**: The width people can use, often less than the clear width because people keep away from walls.
- **Zipper effect**: Overlapping lanes formed by people passing through a bottleneck.
- **Demand**: The rate at which people want to pass a point.

Reading

- Seyfried et al. (2009): New insights into pedestrian flow through bottlenecks: Experiments on bottleneck width and flow. https://arxiv.org/abs/physics/0702004
- Hoogendoorn and Daamen (2005): Pedestrian behavior at bottlenecks: The zipper effect and stepwise capacity. https://doi.org/10.1287/trsc.1040.0102
- Itaewon crowd crush analysis (PLOS ONE 2024): A narrow, sloping alley with flows from both ends. https://doi.org/10.1371/journal.pone.0306764

1.5 Self-organisation: lanes, oscillations and waves

30 minutes · Patterns

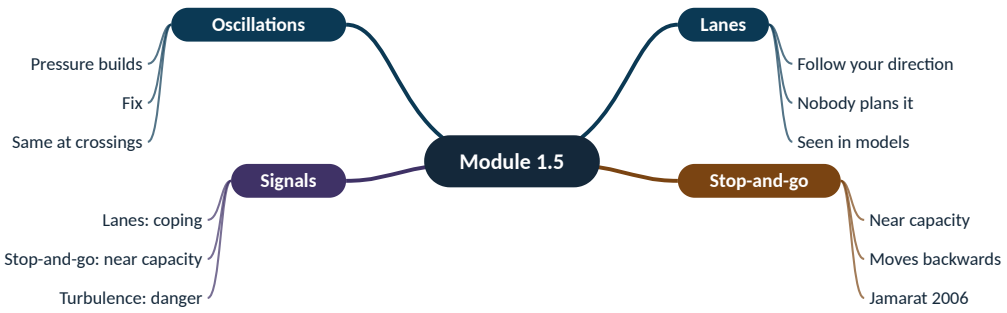
Without a leader, crowds form lanes in counterflow, take turns at narrow doors, and make stop-and-go waves at high density. Simple local rules create these patterns.

By the end you can

- describe lane formation, door oscillations and stop-and-go waves
- explain how simple local rules can create large patterns
- recognise when a pattern helps the flow and when it warns of danger

Overview

Watch a busy station corridor where people walk in both directions. Nobody gives orders, yet within seconds the crowd sorts itself into lanes: people heading the same way follow each other, because walking behind someone going your way means fewer collisions. At a narrow door with people going both ways, small groups pass in turns, like traffic lights nobody installed. In very dense crowds, movement becomes stop-and-go: waves of stopping travel backwards through the crowd, like the ripples in a traffic jam on a motorway. These patterns are called self-organisation. They come from simple, local rules: avoid the person in front, follow someone going your way, step into a gap. Some patterns help, like lanes, which raise flow. Others are warnings: stop-and-go waves mean the crowd is near or above capacity. In this module you will see these patterns emerge in a small simulation, and learn to read them as signals.



Lanes in counterflow

In a two-way corridor, people who follow someone walking their way meet fewer oncoming walkers. This small advantage spreads: soon the corridor carries **lanes of uniform direction**. Lane formation is one of the most robust findings in crowd dynamics and appears in experiments and in models such as the social force model (Helbing and Molnár 1995).

TRY **Look for it**

Next time you walk through a busy station or market, watch for lanes. Count how many there are across the corridor.

Taking turns at doors

At a narrow door with people going both ways, one side wins for a while, then pressure on the other side grows and the direction flips. The result is **oscillation**: alternating bursts. One-way doors, or separate in and out doors, remove the conflict.

Stop-and-go waves

When density is near capacity, one person slowing down makes the person behind slow down a little more, and so on. The disturbance grows and travels **backwards** through the crowd as a stop-and-go wave. Helbing, Johansson and Al-Abideen (2007) filmed such waves at the Jamarat Bridge before the 2006 disaster, followed by crowd turbulence (module 1.7).

MODEL **Like traffic**

Stop-and-go waves in crowds behave much like phantom jams on motorways: no accident needed, just density near capacity.

Reading patterns as signals

Patterns are information. **Lanes** mean the crowd is coping. **Stop-and-go** means it is near capacity. **Irregular, random-looking surges** mean it is far beyond safe density. Control-room staff learn to watch for these changes on cameras, often before any count is available.

LAB **Counting lanes from a snapshot**

Splits the corridor into strips and counts how often the dominant direction changes. With random positions there are many short 'lanes'; organised crowds show few wide ones.

```
lanes: (a number between 5 and 10 for random positions)
```

Takeaways

- Crowds self-organise: lanes, turn-taking and waves emerge from local rules.
- Lanes in counterflow reduce conflicts and raise flow.
- Stop-and-go waves travel backwards and signal density near capacity.
- Patterns are signals for crowd managers.

Check yourself

1. Why do lanes form in two-way corridors?
   (a) Signs tell people to · (b) Following someone going your way means fewer collisions · (c) People walk faster in lanes · (d) Lighting

2. Stop-and-go waves travel…
   (a) Forwards, with the crowd · (b) Backwards through the crowd · (c) Sideways only · (d) Not at all
3. Oscillating flow at a narrow two-way door can be removed by…
   (a) Painting arrows only · (b) Separate doors for each direction · (c) Faster walking · (d) More lighting
4. Which pattern is the most urgent warning?
   (a) Lanes · (b) Zipper effect · (c) Irregular surges in a very dense crowd · (d) People chatting in groups

Answers: 1 (b) A small local advantage spreads into lanes.; 2 (b) Like traffic jams, they move upstream.; 3 (b) Separating directions removes the conflict.; 4 (c) Turbulent motion signals extreme density.

Words

- **Self-organisation**: Large-scale patterns that arise from local interactions without central control.
- **Lane formation**: Sorting of a two-way flow into lanes of uniform direction.
- **Oscillation**: Alternating passage of opposite flows through a narrow opening.
- **Stop-and-go wave**: A wave of stopping that travels backwards through a dense crowd.

Reading

- Helbing and Molnár (1995): Social force model: The model that reproduced lanes and oscillations. https://doi.org/10.1103/PhysRevE.51.4282
- Helbing, Johansson and Al-Abideen (2007): Stop-and-go waves and turbulence at the Jamarat Bridge. https://arxiv.org/abs/physics/0701203
- Moussaïd, Helbing and Theraulaz (2011): How simple rules determine pedestrian behavior and crowd disasters: Simple heuristics that produce lanes and turbulence. https://doi.org/10.1073/pnas.1016507108

1.6 People, not particles: behaviour and choice

35 minutes · Behaviour

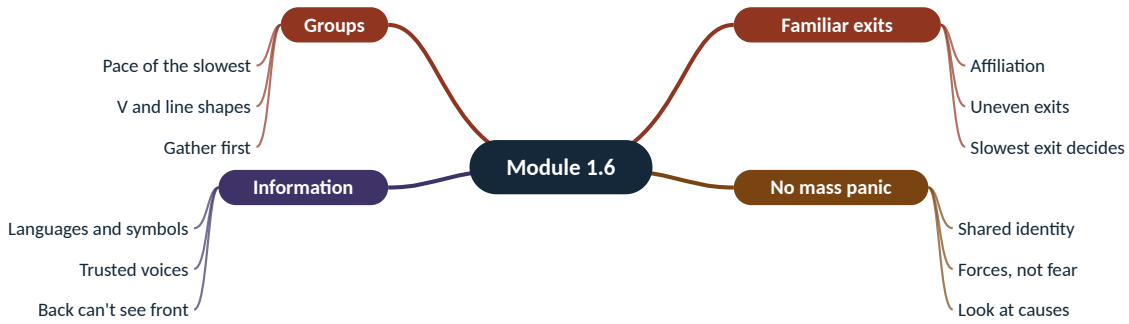
People choose routes, walk in groups, follow familiar paths and respond to information. Crowd behaviour is social, and 'panic' is rare. Good management works with that behaviour.

By the end you can

- explain why people often use familiar routes even when others are free
- describe how groups and social identity shape crowd behaviour
- explain why information, not just space, changes how crowds move
- challenge the myth of mass panic with evidence

Overview

Physics explains a lot about crowds, but people are not particles. They have goals, habits and friends. When people leave a building, many head for the door they came in by, even when another exit is closer and empty. Families and friends walk together, and a group moves at the pace of its slowest member. People look at what others do and follow them. In emergencies, research shows that people mostly stay calm and help each other, partly because they come to feel part of a shared group. Real problems come from missing or confusing information, delays in deciding to move, and designs that send everyone the same way. That is why good crowd management is not only about widths and capacities. It is about signs that people understand, announcements they trust, stewards they can ask, and routes that match what people naturally do. In this module you will explore how behaviour changes evacuation times, and why the myth of mass panic is so persistent and so misleading.



Familiar routes

Studies of evacuations and exit choice show that many people head for the exit they know, often the one they entered by, even if another is nearer and free. Sime (1985) called this **affiliation** with familiar places and people. The result: one exit queues while another stands empty.

$$\text{evacuation time} \approx \text{time of the slowest exit to empty}$$

TRY Explore it

Use the explorer: with 80% of people heading for the main entrance, how much longer does the evacuation take than with an even split?

Groups

At events, most people come with family or friends. Groups move together, walk more slowly, and form walking shapes such as a line abreast or a V (Moussaïd et al. 2010). In evacuations, people look for their group before leaving, which adds to **pre-movement time**.

The myth of panic

Films and headlines show crowds 'panicking'. Research on real emergencies finds the opposite: people are mostly orderly and helpful, and selfish behaviour is rare (Drury, Cocking and Reicher 2009). What looks like panic in a crush is people being pushed by forces they cannot resist. Blaming panic draws attention away from design, information and management.

Information changes movement

People decide using what they see and hear. Clear signs in the crowd's languages, announcements from trusted voices, and visible stewards shorten decision times and balance routes. Poor information, or none, is a cause in many incidents: crowds keep pushing forward because those at the back cannot see what is happening at the front.

RESEARCH Pilgrims and information

At the Hajj, pilgrims come from over 150 countries and speak many languages. Group leaders, colour-coded signs and staggered schedules are part of how information reaches them.

LAB Familiar exit bias

Total time is set by the slower exit. Try changing the walking times.

```
50% at the main entrance: everyone out in 3.8 min
...
90% at the main entrance: everyone out in 6.3 min
```

Takeaways

- People often choose familiar routes, which leaves some exits underused.
- Groups move together and slow evacuations.
- Mass panic is a myth; cooperation is the norm, and crushes are physical.
- Clear, early, trusted information changes how crowds move.

Check yourself

1. Why do many people use the entrance they came in by during an evacuation?
   (a) It is always closest · (b) Familiarity and affiliation · (c) Signs require it · (d) It is the widest

2. What does research say about behaviour in emergencies?
   (a) Mass panic is common · (b) People mostly cooperate and help · (c) People freeze · (d) Everyone runs
3. How do groups affect evacuations?
   (a) They speed them up · (b) They slow them, as people gather and walk together · (c) No effect · (d) They always use the nearest exit
4. Why do people at the back of a crowd keep pushing forward?
   (a) They want to hurt others · (b) They cannot see what is happening at the front · (c) They are told to · (d) To keep warm

Answers: 1 (b) People prefer known routes (Sime 1985).; 2 (b) Drury and colleagues found cooperation is the norm.; 3 (b) Gathering adds to pre-movement time.; 4 (b) Missing information is a common cause.

Words

- **Affiliation**: The tendency to move towards familiar people and places (Sime 1985).
- **Pre-movement time**: Time between an alarm and the start of purposeful movement.
- **Social identity**: A sense of belonging to a group, which can grow among strangers in an emergency.
- **Mass panic**: The myth that crowds in danger turn irrational and selfish.

Reading

- Drury, Cocking and Reicher (2009): Shared identity and cooperation in emergencies. https://doi.org/10.1348/014466608X357893
- Moussaïd et al. (2010): The walking behaviour of pedestrian social groups: Group walking shapes and speeds. https://doi.org/10.1371/journal.pone.0010047
- Sieben, Schumann and Seyfried (2017): Collective phenomena in crowds: Why pedestrian dynamics needs social psychology. https://doi.org/10.1371/journal.pone.0177328

1.7 When crowds become dangerous

40 minutes · Risk

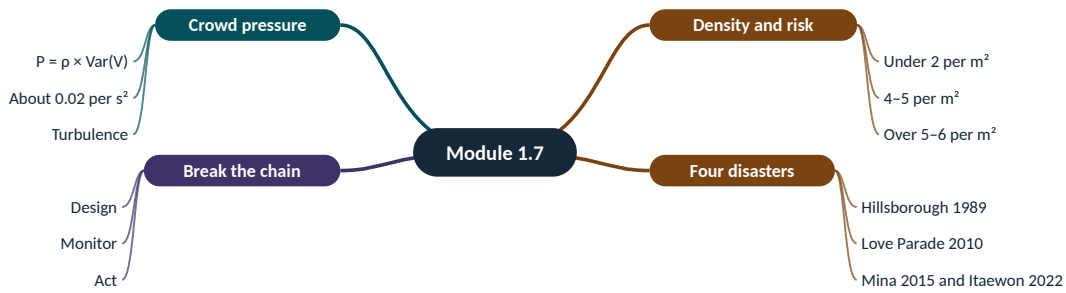
Risk rises steeply with density. Above about 5 to 6 people per m² people lose control of their movement; forces add up through bodies, and crowd turbulence and collapse can follow.

By the end you can

- describe how risk changes with density
- explain crowd pressure and crowd turbulence
- describe the causal chain of a crowd crush
- use the lessons of Hillsborough, Love Parade, Mina and Itaewon

Overview

Most crowds are safe. Danger comes when density gets very high. Above about five or six people per square metre, nobody can move independently: you are carried by the crowd. Forces from many people pushing, even gently, add up through bodies and can reach levels that make breathing impossible. Researchers Dirk Helbing, Anders Johansson and Habib Zein Al-Abideen studied video of the 2006 Jamarat Bridge disaster during the Hajj. They saw the crowd change from smooth flow to stop-and-go waves, and then to crowd turbulence: irregular, random-looking surges that pushed people several metres and made them fall. They measured crowd pressure, density times the variance of local speeds, and found it rose sharply before people fell. Once one person falls in such a crowd, others fall over them, and a crowd collapse spreads. The lesson from Hillsborough, Love Parade, Mina and Itaewon is the same: keep density well below critical levels, watch for warning signs, and never let large flows meet in narrow spaces.



Risk rises steeply with density

Below about 2 people per m² risk is low. Around **4 to 5 per m²** people are in frequent contact. Above about **5 to 6 per m²** they cannot move independently, and forces from others can be transmitted through them (Fruin 1993; Still 2014). Densities of 7 to 9 per m² or more have been measured in fatal incidents.

Crowd pressure and turbulence

Helbing, Johansson and Al-Abideen (2007) analysed video of the Jamarat Bridge entrance before the January 2006 disaster. Flow changed from laminar to **stop-and-go**, then to **crowd turbulence**. They defined crowd pressure as:

$$P = \rho \times \mathrm{Var}(V) \quad (\text{density} \times \text{variance of local velocity})$$

In their data, falls and deaths followed when P rose above about 0.02 per s². Pressure is high only when the crowd is both dense and moving irregularly, which makes it a useful warning signal.

TRAP One number is not a law

The 0.02 s^{-2} value comes from one site and one camera set-up. It is a research finding, not a design standard.

Four disasters, one lesson

Hillsborough, Sheffield (1989): 97 deaths after a crowd was let into already full central pens. **Love Parade, Duisburg (2010)**: 21 deaths on a ramp that was the only way in and out. **Mina, during the Hajj (2015)**: two large flows met at a junction of streets 204 and 223; the official Saudi toll was 769, independent counts exceed 2,000. **Itaewon, Seoul (2022)**: 159 deaths in a sloping alley about 3 to 4 m wide, fed from both ends with no event plan.

In each case, flows were allowed to meet or accumulate in a confined space faster than it could empty.

RESEARCH Go 3 returns here

Module 3.2 analyses these disasters in depth with the DIM-ICE model and the published evidence.

Intervene early

The chain has many links, and the early ones are easiest to break. **Before the event**: design routes so flows do not meet head-on, size spaces for the peak, plan for the unexpected. **During the event**: watch density and warning signs, stop or meter inflow early, open relief routes, and tell people at the back what is happening.

LAB Crowd pressure from local speeds

Same density, very different pressure. Speed variance is what turns a dense crowd into a dangerous one.

```
calm     : variance 0.0001, pressure 0.001 per s2
turbulent: variance 0.0244, pressure 0.171 per s2 -> WARNING
```

Takeaways

- Above about 5–6 per m² people lose control of their movement.
- Crowd pressure = density × variance of local velocity; high values preceded falls at Jamarat in 2006.
- Crushes follow a chain: demand above capacity, high density, forces, turbulence, falls.
- The disasters at Hillsborough, Love Parade, Mina and Itaewon share one cause: flows meeting or accumulating in confined spaces.

Check yourself

1. Above roughly what density can people no longer move independently?
   (a) 1 per m² · (b) 2 per m² · (c) 5–6 per m² · (d) 20 per m²

2. Crowd pressure as defined by Helbing and colleagues is…
   (a) Density × variance of local velocity · (b) Flow × width · (c) Speed squared · (d) Number of people
3. What do most crush victims die of?
   (a) Trampling · (b) Compressive asphyxia · (c) Heat stroke · (d) Heart attacks only
4. A common factor in Mina 2015 and Itaewon 2022 was…
   (a) Fire · (b) Large flows meeting in narrow streets · (c) Bad weather · (d) Too many stewards

Answers: 1 (c) Above about 5 to 6 per m^2 control is lost.; 2 (a) $P = \rho \times \mathrm{Var}(V)$.; 3 (b) They cannot breathe.; 4 (b) Meeting flows in confined spaces.

Words

- **Crowd pressure**: Local density × variance of local velocity; a warning indicator (Helbing et al. 2007).
- **Crowd turbulence**: Irregular, random-looking motion of a very dense crowd that displaces people and causes falls.
- **Crowd collapse**: Progressive falling of people in a dense crowd.
- **Crowd crush**: An incident in which people are compressed so much that they are injured or killed.

Reading

- Helbing, Johansson and Al-Abideen (2007): The dynamics of crowd disasters: Crowd pressure and turbulence at the Jamarat Bridge. https://doi.org/10.1103/PhysRevE.75.046109
- Helbing and Mukerji (2012): Crowd disasters as systemic failures (Love Parade): A detailed open-access analysis of the Love Parade disaster. https://doi.org/10.1140/epjds7
- Hillsborough Independent Panel report (2012): The official review of the 1989 disaster. https://www.gov.uk/government/publications/the-report-of-the-hillsborough-independent-panel

1.8 Planning and managing safe crowds

40 minutes · Practice

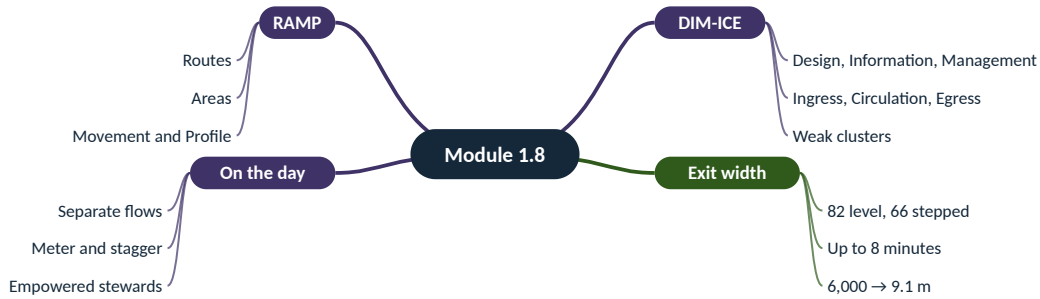
Safe events are designed, informed and managed in every phase. DIM-ICE and RAMP structure the risk analysis; flow rates and egress times size the exits.

By the end you can

- use DIM-ICE to structure a crowd risk assessment
- describe RAMP analysis: routes, areas, movement and profile
- size an exit with a flow rate and an egress time
- list practical measures that keep density down

Overview

How do organisers keep crowds safe? Crowd scientist Keith Still teaches two tools. The first, DIM-ICE, crosses three things organisers control, design, information and management, with the three phases of an event, ingress, circulation and egress. Each of the nine cells is a question: is the design of the entry route adequate, does information reach people during circulation, how is egress managed? The second tool, RAMP, looks at routes, areas, movement over time, and the profile of the crowd. Then come numbers. Guidance for sports grounds in the United Kingdom, the Green Guide, gives planning flow rates of eighty two people per metre width per minute on level routes and sixty six on stepped routes, and egress times of up to eight minutes. Divide the number of people by the flow rate and the time, and you get the exit width needed. Finally, practical measures keep density down: separate flows, one-way systems, holding areas, staggered times, and stewards with clear authority to act.



DIM-ICE

Keith Still's **DIM-ICE** model crosses what organisers control with the phases of an event:

	Ingress	Circulation	Egress
Design	entry routes, gates	concourses, toilets	exits, routes away

	Ingress	Circulation	Egress
Information	tickets, signs	screens, announcements	exit signs, PA
Management	queues, searches	stewarding	staggered release

Each cell is a question to ask. Clusters of weak cells point to where an incident is likely.

RAMP analysis

RAMP asks about **Routes** (where people come from and go), **Areas** (holding and moving spaces, measured), **Movement** (arrival and departure profiles over time) and **Profile** (who the crowd is: age, culture, familiarity, purpose). A football crowd, a family festival and a pilgrimage need different plans even in the same space.

Sizing exits with flow rates

The Green Guide (6th edition) gives planning flow rates of **82 people per metre width per minute** on level routes and **66** on stepped routes, and egress times of up to **8 minutes** (less where fire risk is higher).

$$\text{exit width} = \text{people} \div (\text{flow rate} \times \text{egress time})$$

Example: 6,000 people, level route, 8 minutes: 6,000 ÷ (82 × 8) ≈ **9.1 m**. Stepped route: 6,000 ÷ (66 × 8) ≈ 11.4 m. Note these are per minute, unlike the per-second specific flows of module 1.4.

TRAP Rates are not laws of nature

Planning rates include safety margins and apply to the venues they were written for. Use the guidance that applies where you work, and check routes beyond the exit too.

Keeping density down on the day

Good plans fail without good operations. Measures that work: **separate in and out flows**; **one-way systems**; **holding areas** and **metering** to release people in batches; **staggered** start and finish times; clear **information** in the crowd's languages; and **stewards** with the authority to stop inflow when density rises.

TRY Plan it

A stand of 4,000 needs to leave in 8 minutes by a stepped route. Width? *4,000 ÷ (66 × 8) ≈ 7.6 m.*

LAB Exit width calculator

Width = people ÷ (rate × time). Change the time to 6 minutes for a higher fire risk.

```
2000 people: level 3.0 m, stepped 3.8 m (8 min)
6000 people: level 9.1 m, stepped 11.4 m (8 min)
```

Takeaways

- DIM-ICE crosses Design, Information and Management with Ingress, Circulation and Egress.
- RAMP analysis looks at Routes, Areas, Movement and Profile.

- Exit width = people ÷ (flow rate × egress time); the Green Guide uses 82 (level) and 66 (stepped) per metre per minute and up to 8 minutes.
- On the day, separate flows, metering, information and empowered stewards keep density down.

Check yourself

1. In DIM-ICE, the 'I' in DIM stands for…
   (a) Ingress · (b) Information · (c) Injury · (d) Inspection
2. 6,000 people must leave in 8 minutes by a level route at 82 per m per min. Width?
   (a) 4.6 m · (b) 9.1 m · (c) 18 m · (d) 73 m
3. RAMP stands for…
   (a) Routes, Areas, Movement, Profile · (b) Risk, Assessment, Management, Plan · (c) Ramps, Aisles, Main exits, Paths · (d) Rate, Area, Mass, Pressure
4. Why are stepped routes given a lower flow rate than level routes?
   (a) Stairs are wider · (b) People move more slowly and carefully on steps · (c) Fewer people use stairs · (d) To save money

Answers: 1 (b) Design, Information, Management; then Ingress, Circulation, Egress.; 2 (b) 6,000 ÷ (82 × 8) ≈ 9.1 m.; 3 (a) Still's RAMP analysis.; 4 (b) Steps slow people down.

Words

- **DIM-ICE**: Still's risk model crossing Design, Information and Management with Ingress, Circulation and Egress.
- **RAMP analysis**: Routes, Areas, Movement and Profile: a structured look at a crowd and its site.
- **Egress time**: The time for people to leave a space or venue.
- **Green Guide**: Guide to Safety at Sports Grounds (UK), giving planning flow rates and egress times.
- **Metering**: Controlling the rate at which people enter a space.

Reading

- Still et al. (2020): Place crowd safety, crowd science? (open access): DIM-ICE applied to real case studies. https://e-space.mmu.ac.uk/625559/
- Sports Grounds Safety Authority: the Green Guide: The Guide to Safety at Sports Grounds, 6th edition. https://sgsa.org.uk/greenguide/
- HSE: Managing crowds safely (HSG154): Planning, stewarding and monitoring crowds. https://www.hse.gov.uk/pubns/books/hsg154.htm

Questions for spaced practice

From the question bank: answer from memory, then check. Revisit after 1, 3, 7 and 14 days.

1.1 What a crowd is, and why it matters

1. **Recall:** Name the three quantities used to describe a moving crowd.
   → Density (people per m²), speed (m/s) and flow (people per second or minute, often per metre of width).
2. **Recall:** What are the three phases of an event?
   → Ingress (arriving), circulation (moving around inside) and egress (leaving).
3. **Calculate:** A gate passes 60 people a minute; 100 a minute arrive for 30 minutes. How many are waiting?
   → (100 − 60) × 30 = 1,200 people.
4. **Explain:** Why does the largest queue form after the peak of arrivals?
   → Because the queue keeps growing as long as arrivals exceed capacity, which continues for a while after the peak; it only shrinks once arrivals fall below capacity.
5. **Explain:** Why is 'panic' a misleading explanation for crowd crushes?
   → Most crush victims die from compressive asphyxia in crowds that are barely moving; research shows people mostly cooperate. The cause is too much density, created by failures of design, information and management.
6. **Apply:** Give two things organisers can change to stop a queue growing at a gate.
   → Increase capacity (open more gates or lanes) or reduce or spread arrivals (staggered entry times, information, metering earlier on the route).

1.2 Space, density and level of service

1. **Recall:** Define density and give its unit.
   → The number of people per unit area, in people per square metre (per m²).
2. **Calculate:** 300 people wait in a 100 m² area. What is the density and the space each?
   → 3 people per m²; 1 ÷ 3 ≈ 0.33 m² each.
3. **Recall:** What does Fruin's level of service A mean on a walkway?
   → More than about 3.25 m² per person: free choice of speed and path.
4. **Explain:** Why do queuing areas use different density thresholds from walkways?
   → Standing people need less space than walking people, so they tolerate higher densities before discomfort; walking needs room for stride and passing.
5. **Calculate:** How many people does a 1,600 m² usable area hold at 2 per m²?
   → 1,600 × 2 = 3,200 people.
6. **Apply:** A report says a square was 'only half full'. What would you ask?
   → Where the people were: was density uneven, with hot spots near stages, screens or exits much higher than the average?

1.3 Speed, flow and the fundamental diagram

1. **Recall:** State the relation between flow, density and speed.
   → Specific flow J = density × speed, in people per metre of width per second; total flow = J × width.
2. **Calculate:** A 3 m wide corridor has density 1.5 per m² and speed 0.8 m/s. What is the total flow?
   → J = 1.5 × 0.8 = 1.2 per m per s; total = 1.2 × 3 = 3.6 people per second.
3. **Explain:** Why does flow peak and then fall as density rises?
   → Flow is density times speed. At low density, adding people raises flow; but speed falls ever faster as people get in each other's way, so beyond a point the product falls.
4. **Recall:** What is capacity in Weidmann's fundamental diagram, and at what density?
   → About 1.22 people per metre per second, at about 1.75 people per m².

5. **Apply:** A planner uses Weidmann's curve for elderly pilgrims. What is the risk?
   → Older or slower crowds have lower free speeds and capacities, so the plan may overestimate flow and underestimate queues and density.
6. **Explain:** Why is a corridor operated near capacity fragile?
   → A small disturbance can push it onto the congested branch, where flow drops and density keeps rising, so it recovers only when demand falls.

1.4 Bottlenecks and capacity

1. **Recall:** What is a bottleneck?
   → The narrowest point on a route, which limits the flow of the whole route.
2. **Calculate:** A 1.5 m door has specific flow 1.3 per m per s. How long do 1,000 people take?
   → Capacity 1.95 per s; 1,000 ÷ 1.95 ≈ 513 s ≈ 8.5 minutes.
3. **Recall:** What range of specific flows did bottleneck experiments find?
   → Roughly 1.3 to 1.9 people per metre per second (Seyfried et al. 2009).
4. **Explain:** Why does widening the corridor before a narrow door not help?
   → The door still sets the capacity; extra width upstream only stores more of the queue.
5. **Calculate:** Arrivals 4 per s, capacity 3 per s. How many are queuing after 15 minutes?
   → (4 − 3) × 900 = 900 people.
6. **Apply:** Name two places where a bottleneck is especially dangerous.
   → At the bottom of a slope or stair, where people above push forward; and where two flows meet head-on or at a T-junction.

1.5 Self-organisation: lanes, oscillations and waves

1. **Recall:** Name three self-organised crowd patterns.
   → Lane formation in counterflow, oscillations at narrow two-way doors, and stop-and-go waves at high density.
2. **Explain:** How do lanes form without anyone planning them?
   → Walking behind someone going your way means fewer collisions; each person makes this small choice and the advantage spreads into lanes.
3. **Explain:** Why are stop-and-go waves a warning sign?
   → They appear when density is near or above capacity, where small slowdowns grow; they can precede crowd turbulence, as at Jamarat in 2006.
4. **Apply:** How would you remove oscillations at a narrow door used in both directions?
   → Make it one-way, for example separate entry and exit doors, or a central divider.
5. **Recall:** In which direction do stop-and-go waves travel?
   → Backwards, against the walking direction.
6. **Apply:** A control room sees lanes turn into stop-and-go motion. What should happen?
   → Treat it as a sign of density near capacity: reduce or stop inflow, open relief routes and inform the crowd before turbulence develops.

1.6 People, not particles: behaviour and choice

1. **Recall:** What is affiliation in exit choice?
   → The tendency to move towards familiar places and people, so people often leave by the route they entered.
2. **Explain:** Why is 'mass panic' a poor explanation of crowd disasters?
   → Evidence shows people mostly cooperate; crush deaths come from physical density and pressure caused by design, information and management failures.
3. **Calculate:** 1,200 people, two exits of 3 per s each, 80% use one. Roughly when is the busy exit empty?
   → 960 ÷ 3 = 320 s plus about 20 s walking ≈ 5.7 minutes.

4. **Recall:** Name two ways groups change crowd movement.
   → Groups walk at the pace of their slowest member, form shapes such as lines or Vs, and gather before evacuating, adding pre-movement time.
5. **Apply:** How could you improve information for a multilingual crowd?
   → Use pictograms and colour codes, signs in the main languages, group leaders who brief their groups, and visible stewards.
6. **Explain:** Why do people at the back push forward even when the front is crushed?
   → They cannot see or hear what is happening at the front, so they keep moving towards their goal.

1.7 When crowds become dangerous

1. **Recall:** At what density do people lose the ability to move independently?
   → Above about 5 to 6 people per m^2.
2. **Recall:** Define crowd pressure.
   → Local density multiplied by the variance of local velocity, $P = \rho \times \mathrm{Var}(V)$ (Helbing et al. 2007).
3. **Explain:** Describe the chain of events leading to a crowd crush.
   → Demand exceeds capacity, density rises, people lose control and forces add up through bodies, turbulence pushes people over, falls spread into a crowd collapse, and people die of compressive asphyxia.
4. **Recall:** What happened at the Love Parade in 2010?
   → In Duisburg, 21 people died at a ramp that was the only way in and out of the festival area, where incoming and outgoing flows met.
5. **Apply:** Name three warning signs control-room staff can watch for.
   → Stop-and-go waves, swaying surges or turbulence, people climbing out over barriers, and distress or emergency calls.
6. **Explain:** Why is the 0.02 per s^2 pressure value not a design standard?
   → It comes from one site's video data and measurement method; values depend on how density and velocity are measured, so it is a research indicator, not a universal limit.

1.8 Planning and managing safe crowds

1. **Recall:** What does DIM-ICE stand for?
   → Design, Information, Management crossed with Ingress, Circulation, Egress.
2. **Recall:** What does RAMP stand for?
   → Routes, Areas, Movement, Profile.
3. **Calculate:** 4,000 people, stepped route, 8 minutes. Exit width?
   → $4{,}000 \div (66 \times 8) \approx 7.6$ m.
4. **Explain:** Why does the same space need different plans for different crowds?
   → Profiles differ: age, mobility, culture, familiarity, purpose and group size change speeds, routes and behaviour.
5. **Apply:** Name four measures that keep density down on the day.
   → Separating in and out flows, one-way systems, holding areas and metering, staggered times, clear multilingual information, and empowered stewards.
6. **Explain:** Why is egress often the hardest phase?
   → Everyone wants to leave at once at the end, so the peak flow is concentrated in a short time, often with tired people and darkness.